

WFI SERIES (1 TO 30 TONS/DAY)

Heavy-Duty Industrial Ice Flake Machines

Continuous Production of Sub-Cooled Dry Ice Flakes (-5°C to -8°C) with Stationary Vertical SUS304 Evaporator Drums & Bitzer Compressors



1-30 TPD

Daily Ice Production
Capacity (1,000 to
30,000 kg)

-5°C TO -8°C

Subcooled, Dry, Non-
Clumping Flakes (1.5-2.2
mm)

STATIONARY DRUM

Zero Dynamic Seals
Prevents Refrigerant
Leaks

SUS304 DRUM

Food-Grade Stainless
Steel Wetted Evaporator
Wall

THERMODYNAMIC ARCHITECTURE & ENGINEERING

Operational Overview & Engineering Advantages

Continuous industrial duty engineered for 45°C tropical ambient conditions

Win Equipments WFI Series industrial ice flake machines produce continuous, sub-cooled dry ice flakes engineered for commercial fisheries, poultry processing, concrete dam pouring, and chemical synthesis. Unlike conventional rotating-drum ice machines that suffer from refrigerant seal leaks, the WFI Series employs a stationary vertical SUS304 evaporator drum with an internal rotating spiral ice blade, delivering zero refrigerant leakage, ultra-low maintenance, and maximum thermodynamic efficiency.

Working Principle: Fresh or seawater is pumped from a lower stainless steel basin into an overhead rotary water distributing pan that coats the inner vertical wall of the stationary evaporator drum with an even, continuous film. Low-temperature eco-friendly refrigerant boiling within the drum jacket freezes the water instantly on the SUS304 inner surface. A high-strength stainless steel helical scraper rotates slowly, peeling sub-cooled dry ice flakes (-5°C to -8°C, 1.5 to 2.2 mm thick) from the wall. The dry flakes drop freely by gravity through the open drum bottom into storage bins or pneumatic conveying systems with zero clumping.

SYSTEM INTEGRATION & P&ID FLOW ARCHITECTURE

Feed Water → Stainless Basin → Vertical Distribution Pan → Stationary SUS304 Drum Wall (-22°C Evaporation) → Internal Spiral Scraper → Free Fall Dry Flakes (-7°C) → Insulated Storage Vault



Stationary Vertical SUS304 Evaporator Drum

The evaporator drum remains completely stationary while only the internal scraper blade rotates. Eliminates high-pressure rotary mechanical refrigerant seals—the #1 cause of catastrophic refrigerant leaks in rotating-drum machines.



Sub-Cooled, Flat & Edueless Flake Geometry

Produces thin (1.5–2.2 mm) sub-cooled dry flakes with zero free water. Their smooth, rounded edges provide massive surface contact area for rapid thermal exchange without scratching or bruising delicate fish skin or poultry meat.



Heavy-Duty Bitzer / Danfoss Refrigeration Package

Driven by world-renowned Bitzer semi-hermetic reciprocating or screw compressors equipped with liquid injection cooling, oil separators, suction accumulators, and high-efficiency water-cooled shell-and-tube condensers.



Food-Grade Hygienic Sanitary Construction

All surfaces contacting water and ice—evaporator wall, water distribution pan, ice blade, and collection basin—are fabricated from SUS304 food-grade stainless steel compliant with HACCP and USFDA food safety standards.

CERTIFIED ENGINEERING RATINGS

Technical Specifications Matrix

Full standard model range & operational parameters

Standard Design Conditions: Ice Production Basis: Water Inlet +16°C | Ambient Temp +25°C | Evaporation Temp -22°C | Condensing Temp +40°C | Flake Temp: -5°C to -8°C | Flake Thickness: 1.5 to 2.2 mm | Drum: Food Grade SUS304

MODEL CODE	DAILY OUTPUT TPD	DAILY OUTPUT KG/24H	POWER KW	COMPRESSOR MAKE & HP	CONDENSER TYPE	SKID DIMENSIONS (L×W×H MM)	OPER WEIGHT KG
WFI 010	1 TPD	1,000 kg	4.8 kW	Danfoss / Bitzer (5 HP)	Air / Water Cooled	1350 × 1050 × 980	580 kg
WFI 020	2 TPD	2,000 kg	8.5 kW	Bitzer Semi-Herm (8.5 HP)	Water Cooled Shell & Tube	1550 × 1200 × 1150	850 kg
WFI 030	3 TPD	3,000 kg	12.5 kW	Bitzer Semi-Herm (12 HP)	Water Cooled Shell & Tube	1800 × 1350 × 1300	1,150 kg
WFI 050	5 TPD	5,000 kg	19.5 kW	Bitzer Semi-Herm (20 HP)	Water Cooled Shell & Tube	2100 × 1500 × 1550	1,850 kg
WFI 100	10 TPD	10,000 kg	38.0 kW	Bitzer Semi-Herm (40 HP)	Water Cooled Shell & Tube	2600 × 1800 × 1850	3,200 kg
WFI 150	15 TPD	15,000 kg	56.0 kW	Bitzer Screw (60 HP)	Water Cooled Shell & Tube	3100 × 2000 × 2100	4,600 kg
WFI 200	20 TPD	20,000 kg	74.0 kW	Bitzer Screw / Twin Recip	Water Cooled Shell & Tube	3600 × 2200 × 2300	5,800 kg
WFI 300	30 TPD	30,000 kg	110.0 kW	Bitzer / Hanbell Screw	Water Cooled Shell & Tube	4200 × 2400 × 2600	7,500 kg

ENGINEERING SELECTION & OPERATIONAL NOTES

- Seawater on-board marine models manufactured with SUS316L stainless steel drum and marine-grade bronze condensers available on request.
- Complete turnkey ice storage rooms with automatic screw discharge and pneumatic ice rakes custom engineered from 5 to 100 tons storage capacity.
- Refrigerants: R404A / R507A standard; R717 (Ammonia) systems custom engineered for large commercial installations.

FIELD DEPLOYMENT & COMPETITIVE ANALYSIS

Applications & Industrial Benchmarks

Demonstrated operational benefits across high-reliability manufacturing sectors

- Commercial Fisheries & Deep-Sea Trawlers** Cools freshly caught seafood instantly without bruising fish flesh, preserving freshness, texture, and export value for ocean transit.
- Poultry Processing & Meat Abattoirs** Maintains eviscerated carcasses at safe bacteriological temperatures (<4°C) during spin chilling and cutting stages.
- Mass Concrete Cooling for Dams & Highways** Replaces mixing water with dry ice flakes to dissipate the intense heat of hydration, preventing thermal cracking in mass civil structures.
- Chemical & Synthetic Dye Manufacturing** Provides instant thermodynamic heat absorption during violent exothermic diazotization and sulfonation batch reactions.
- Perishable Produce & Supermarket Display** Keeps vegetables, fruits, and fresh catch crisply chilled in retail display cases without dehydration or surface discoloration.

Engineering Comparison: Win WFI vs. Conventional Rotating-Drum Machines		Standard ISO 9001:2015 Quality
Engineering Parameter	Win Equipments Engineered Solution	Standard / Imported Benchmark
Refrigerant Seals	Stationary drum with hard-piped copper/steel joints (0% leak rate)	Dynamic rotating carbon/ceramic seals (Frequent costly leaks)
Drum Metallurgy	Food-grade SUS304 stainless inner wall with polyurethane insulation	Hard-chrome plated carbon steel prone to flaking & rusting
Flake Quality	Dry, sub-cooled (-7°C), flat flakes with zero liquid water	Wet, mushy ice (>10% free water) that clumps in storage
Compressor Quality	Heavy-duty Bitzer / Danfoss compressors with water-cooled condensers	Low-cost semi-hermetic copies with marginal cooling margins

MANUFACTURING STANDARDS & FACTORY VERIFICATION

Factory Quality Assurance & Consultation

Calibrated testing regimes, global certifications, and direct engineering contact

✓ Factory Acceptance Testing (FAT)

- Continuous 48-hour ice production rate and flake thickness FAT verification run
- Helium leak testing of complete evaporator drum jacket and refrigeration piping
- Food contact hygiene swab and surface finish roughness certification (Ra <0.8 µm)
- Overload cut-off and water starvation automatic safety shutdown interlock testing

★ Certified Standards & Accreditations

- ISO 9001:2015 Quality Management System
- IAF (International Accreditation Forum) Accredited
- DAC (Dubai Accreditation Center) Recognized
- Pre-dispatch run-testing under tropical temperature simulation



Manufacturer Warranty Assurance: 24 Months warranty on stationary evaporator drum; 12 months comprehensive on refrigeration unit. Direct field service engineering and genuine replacement spare parts stocked at Arasur Works, Coimbatore.

WIN EQUIPMENTS

SAVE WATER AND POWER • ISO 9001:2015 (IAF & DAC ACCREDITED)

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